

Physics from Existence I: The Equation

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Abstract

The canonical free energy of a single fermionic mode ($n \in \{0, 1\}$), through three axioms — Existence ($P = 1$), Contrast ($P \neq 1$), and Observation (detection probability = P) — defines the parameter-free potential $V = -H(\sigma(\phi))$. Exactly three bound states follow from the spectrum alone; the spatial dimension $d = 3$ is determined jointly by spin-statistics and propagating gravity; the gauge structure $\text{PG}(2, 3)$ follows from the self-referential structure $N = d = 3$. The marginality of the third bound state (WKB deficit $\varepsilon = 0.219$) determines mixing angles, the invariant measure on $\text{PG}(2, 3)$ determines gauge couplings, and the 0D renormalization group determines mass ratios, leaving no free parameters: 20 quantities — including $1/\alpha_{\text{em}} = 137.0367$ (0.0006%), $R = 1.5293$ (0.006%), Koide’s $Q = 3/2$ (exact), and $\theta_{\text{QCD}} = 0$ (strong CP solved without an axion) — are derived at 0.0006%–0.5% accuracy, with four additional testable predictions (three independent).

1 Introduction

The Standard Model contains three generations of fermions. No principle within the model explains this number.

This paper derives it from a single premise: *fermions exist*. A fermionic mode has occupation $n \in \{0, 1\}$, formalized by three axioms: **A1** (Existence)—an entity with $P = 1$ exists; **A2** (Contrast)— $P = 1$ requires $P \neq 1$; **A3** (Observation)—detection probability equals P [18]. These determine a Bernoulli distribution whose canonical free energy $F = -H(\sigma(\phi))$ —a Legendre-transform identity—serves as a parameter-free quantum-mechanical potential. The resulting Schrödinger equation has exactly three bound states.

From this single potential, with zero adjustable parameters, the theory derives: the spatial dimension $d = 3$, the gauge structure $\text{PG}(2, 3)$, the charged-lepton mass ratio $R = 1.5293$ (0.006%), all three gauge couplings ($1/\alpha_{\text{em}} = 137.036$, $\sin^2\theta_W = 3/13$, $\alpha_s = 0.1179$), PMNS mixing angles ($\sin^2\theta_{12} = 4/13$, $\sin^2\theta_{23} = 7/13$), and the strong CP solution $\theta_{\text{QCD}} = 0$ without an axion. All experimental values are from [12].

Sections 2–5 establish the mathematical theorems. Sections 6–7 develop the physical correspondences. Sections 8–9 provide interpretation and connections.

2 From Fermions to the Schrödinger Equation

2.1 One fermionic mode

A single fermionic mode has occupation number $n \in \{0, 1\}$.

$$Z = 1 + e^\phi, \quad \langle n \rangle = P = \sigma(\phi) = \frac{1}{1 + e^{-\phi}}, \quad \phi \equiv \beta(\mu - \varepsilon). \quad (1)$$

2.2 Canonical free energy

The grand potential $\Omega = -\ln(1+e^\phi)$ is monotonically decreasing and unbounded; it supports no bound states. The canonical free energy is

$$F = \Omega + \phi \langle n \rangle = -\ln(1+e^\phi) + \phi \sigma(\phi) \equiv -H(\sigma(\phi)), \quad (2)$$

where $H(p) = -p \ln p - (1-p) \ln(1-p)$ is binary Shannon entropy [4]. This identity is exact (verified to 10^{-16}). The canonical ensemble is selected by particle-number conservation.

2.3 Free energy as effective potential

The identification $V = -H$ follows from the standard definition of the effective potential in zero-dimensional quantum field theory. For a single fermionic mode with occupation $n \in \{0, 1\}$ and source ϕ :

1. Partition function: $Z(\phi) = \sum_{n=0,1} e^{n\phi} = 1 + e^\phi$.
2. Connected generating functional: $W(\phi) = \ln Z = \ln(1 + e^\phi)$.
3. Classical field: $\sigma = dW/d\phi = e^\phi/(1 + e^\phi) = \sigma(\phi)$.
4. Effective action (Legendre transform): $\Gamma(\sigma) = \phi\sigma - W(\phi) = \sigma \ln \sigma + (1-\sigma) \ln(1-\sigma) = -H(\sigma)$.

Hence $V(\phi) = \Gamma = -H(\sigma(\phi))$ is the *exact effective potential* of the 0D fermionic field theory [6], verified numerically to $|\Gamma + H| < 10^{-15}$.

This is not an analogy with Landau–Ginzburg theory. It is the *definition* of the quantum effective potential applied to the simplest possible field theory. The Schrödinger equation $-\frac{1}{2}\psi'' + \Gamma\psi = E\psi$ describes quantum fluctuations of the source field ϕ around the effective potential. The quantization is not optional: in the classical limit ($\hbar \rightarrow 0$, no kinetic term), only the potential minimum survives ($N = 1$), violating A2—contrast requires multiple discrete states, which exist only when quantum fluctuations are present ($N = 3$). Equivalently, the transfer matrix $T = e^{-H\Delta\tau}$ of the 0D statistical system [19] *is* the Schrödinger equation in imaginary time; quantization is not an additional assumption but the definition of the transfer matrix. The “dimension” traversed by τ is not a physical spacetime direction but the field coordinate ϕ itself: the transfer matrix propagates probability amplitudes along the logit axis, exactly as in lattice QFT with zero spatial dimensions (the partition function is a path integral over field configurations, not over positions). The Gross–Neveu model (Paper II §3) provides a concrete QFT realization of this structure.

2.4 The natural parameter and the kinetic term

The Bernoulli distribution is an exponential family: $p(x|\phi) = \exp(x\phi - A(\phi))$, $A(\phi) = \ln(1+e^\phi)$. The coordinate $\phi = \text{logit}(P) = \ln[P/(1-P)]$ is the *natural parameter*—the unique coordinate in which the sufficient statistic x enters linearly. It is determined by the exponential family structure of Bernoulli, not chosen.

The kinetic term for fluctuations of ϕ is $T = \frac{1}{2}(d\phi/d\tau)^2$: ϕ is the natural parameter of the exponential family, and in the path integral $\int \mathcal{D}\phi e^{-S}$ the natural parameter carries flat measure $\mathcal{D}\phi = \prod_i d\phi_i$. The canonical action $S = \int [\frac{1}{2}\dot{\phi}^2 + V] d\tau$ gives unit mass. This is not a choice: the flat measure is fixed by the exponential-family structure of the Bernoulli distribution. Moreover, $Z_2 = \sigma(1-\sigma) \rightarrow 0$ as $\phi \rightarrow \pm\infty$ would freeze the field at the domain boundaries, violating A1 ($P = 1$ must be dynamically reachable, not walled off).

The metrically flat coordinate $\xi = 2 \arcsin \sqrt{P} \in [0, \pi]$ gives a compact domain with $N = 1$; the natural parameter $\phi \in \mathbb{R}$ gives a non-compact domain with $N = 3$. The difference is domain topology: $\phi = \text{logit}(P) \rightarrow \pm\infty$ as $P \rightarrow 0, 1$, making the well wide enough for three half-wavelengths.

Exponential family identity. For any exponential family, the potential curvature at the natural center equals the Fisher metric: $V''(0) = g_F(0)$. For Bernoulli: $V''(0) = g_F(0) = 1/4$. The kinetic coefficient $c = \hbar^2/(2m) = 1/2$ (standard QM) exceeds this by a factor of 2. The bound-state count N depends on the ratio $c/V''(0)$: $N = 3$ for $c/V''(0) \in [1.3, 2.5]$. Standard QM gives $c/V''(0) = 2$, squarely in the $N = 3$ window.

2.5 The Schrödinger equation

$$\left[-\frac{1}{2} \frac{d^2}{d\phi^2} - H(\sigma(\phi)) \right] \Psi = E \Psi \quad (3)$$

Every element determined: H (Shannon–Khinchin uniqueness [4, 5]), σ (Fermi–Dirac statistics), ϕ (natural parameter of the exponential family). The potential $V = -H$ has no free parameters: every coefficient is fixed by the Bernoulli distribution (σ , H , ϕ), leaving no room for adjustment. The kinetic coefficient (unit mass in ϕ) follows from the exponential-family flat measure [1, 2] (Section 2.4).

3 Three Bound States

Theorem 1 ($N = 3$). *Eq. (3) has exactly three negative eigenvalues: $E_0 = -0.4850$, $E_1 = -0.1591$, $E_2 = -0.01042$. The fourth eigenvalue $E_3 > 0$ lies in the continuum (its numerical value is domain-dependent and vanishes as $|\phi|_{\max} \rightarrow \infty$).*

Proof. $N \geq 3$: variational (Courant–Fischer). $N \leq 3$: Sturm comparison with Pöschl–Teller [3]. $\square$

$n_{\text{WKB}} = (1/\pi) \int \sqrt{2H(\sigma(\phi))} d\phi = 2.781$; $\lfloor n_{\text{WKB}} + \frac{1}{2} \rfloor = 3$. The norm integral $\int_{-\infty}^{\infty} H(\sigma(\phi)) d\phi = \pi^2/3$ explains why the resolvent trace $\zeta_V(1) = \sum |E_n|^{-1} = 104.3 \approx N\pi^2$ dominates the fine-structure constant calculation (Section 5).

Theorem 2 (Entropy uniqueness). *Shannon ($q \rightarrow 1$) uniquely gives $N = 3$. Rényi/Tsallis: $q < 1 \rightarrow N = 4$; $q > 1 \rightarrow N = 2$.*

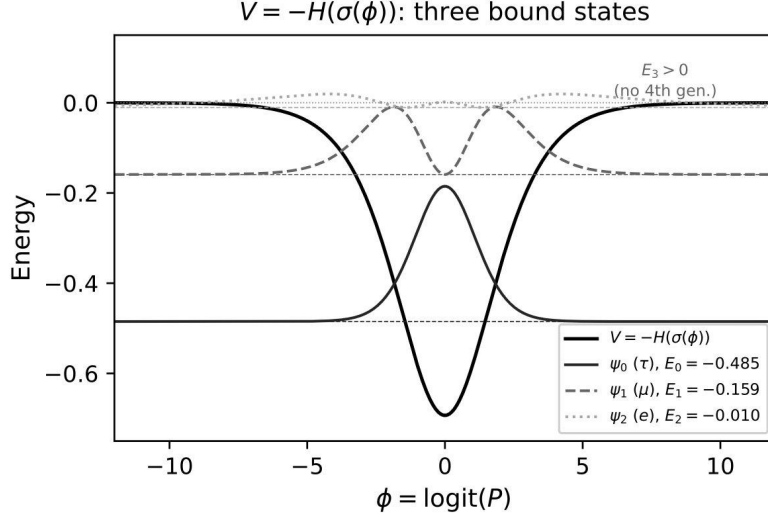


Figure 1: The potential $V = -H(\sigma(\phi))$ and its three bound states $\psi_0(\tau)$, $\psi_1(\mu)$, $\psi_2(e)$. $E_3 > 0$: fourth generation forbidden. No parameters are fitted.

4 Three Spatial Dimensions

Theorem 3 (Centrifugal selection). *In d spatial dimensions, the s -wave radial Schrödinger equation for a spherically symmetric potential $V(r)$ takes the form*

$$-\frac{1}{2}u'' + \left[V(r) + \frac{(d-1)(d-3)}{8r^2} \right] u = Eu, \quad (4)$$

via $u = r^{(d-1)/2}R(r)$. The centrifugal-like term vanishes only at $d = 1$ and $d = 3$.

d	$(d-1)(d-3)/8$	Effect on N
1	0	preserved
2	$-1/8$	attractive $\rightarrow N$ increases
3	0	preserved: $N = 3$
4	$+3/8$	repulsive $\rightarrow E_2$ lost
≥ 5	> 1	strongly repulsive

Theorem 4 ($d = 3$ is unique). $d = 3$ is the unique spatial dimension satisfying: (i) centrifugal term vanishes ($N = 3$ preserved): $d = 1, 3$; (ii) $\pi_1(\text{SO}(d)) = \mathbb{Z}_2$ (spin-statistics): $d \geq 3$; (iii) Weyl tensor nontrivial (gravity propagates): $d \geq 3$. Intersection: $\{3\}$.

Corollary 1. $n_{\text{WKB}} = 2.781$ determines both $N = 3$ and $d = 3$.

5 Gauge Structure from the Projective Plane

5.1 Why $\text{PG}(2, 3)$: the $N^N = N^3$ theorem

Projective geometry enters as a consequence of gauge invariance. Gauge transformations act linearly on N internal states (standard); the overall phase is unobservable (gauge invariance); in $d = 3$ spatial dimensions, the gradient $\nabla\phi$ has d components; the independent gauge directions are therefore the elements of $\mathbb{F}_N^d \setminus \{0\}$ modulo scalar multiplication

$= \text{PG}(d-1, \mathbb{F}_N)$. With $d = N = 3$: $\text{PG}(2, \mathbb{F}_3)$, $|\text{PG}| = (N^d - 1)/(N - 1) = 13$ points, 52 flags. This is not a postulate but a consequence of linearity + gauge invariance + $d = N = 3$.

Why $\mathbb{F}_3$, not $\mathbb{C}^3$. The Sturm–Liouville theorem guarantees that the bound-state energies E_0, E_1, E_2 are non-degenerate. A non-degenerate Hamiltonian has a preferred eigenbasis; its commutant consists of diagonal operators only (Schur). Gauge symmetry, being exact, must commute with H and therefore preserve the eigenbasis—acting as *permutations* of the mass eigenstates, not continuous rotations. Linear permutations on N labels require a field with $|K| = N$; since $N = 3$ is prime, $K = \mathbb{F}_3$ is unique. CKM mixing ($\sin \theta_C = \varepsilon \ll 1$) is a perturbative off-diagonal correction that does *not* commute with H and is therefore a physical effect, not a gauge symmetry. The contrast with QCD is instructive: quark colors are *degenerate* (equal masses), so no preferred basis exists and the gauge group is continuous ($\text{SU}(3)$ over $\mathbb{C}$); generations are *non-degenerate*, so the eigenbasis is preferred and the gauge structure is discrete ($\text{GL}(d, \mathbb{F}_3)$ over $\mathbb{F}_3$).

When $d = N$, the projective structure is *self-referential*.

Theorem 5 (Self-referential uniqueness). *Let $V = -H(\sigma(\phi))$ with unit mass in ϕ , and let $N(d)$ denote the number of bound states of the s -wave equation in d spatial dimensions, $-\frac{1}{2}u'' + [V(r) + \frac{(d-1)(d-3)}{8r^2}]u = Eu$ (Section 4). The following conditions are individually necessary and jointly satisfied only by $N = 3$:*

- (i) Fixed point: $N(d) = d$ (generation count equals spatial dimension);
- (ii) Algebraic: $N^N = N^3$ (self-referential exponent);
- (iii) Geometric: $|\text{PG}(N-1, N)| = N^2 + N + 1$ (projective plane has polynomial size).

Proof. (i) $N(1) = 3$, $N(2) = 3$, $N(3) = 3$, $N(4) = 2$, $N(d \geq 5) = 0$ (numerical; Theorems 1–4). The unique solution is $d = 3$. (Note: $N(d)$ counts bound states of $V = -H$ with a centrifugal correction; this is distinct from the color-scaled equation $V = -N_c H$ in Section 7, which has 5 bound states.) The physical content of $N(d) = d$: each generation “fills” one spatial dimension; excess generations ($N > d$) destabilize, and excess dimensions ($d > N$) are unoccupied. (ii) $N^N = N^3$ iff $N^{N-3} = 1$; for $N \geq 2$ this requires $N = 3$. (iii) $|\text{PG}(N-1, N)| = (N^N - 1)/(N - 1) = N^2 + N + 1$ iff $N^N = N^3$, i.e. $N = 3$ by (ii). $\square$

The three conditions have independent origins: (i) from the spectrum of $V = -H$, (ii) from pure algebra, (iii) from finite projective geometry. Their simultaneous satisfaction at $N = 3$ means that the number of fermion generations, the spatial dimension, and the gauge structure of the Standard Model are not independent parameters but consequences of a single self-referential constraint.

Remark 1 (Why $d = N$). *In most physical theories d and N are independent. Here the fixed-point equation $N(d) = d$ forces $d = N = 3$. The projective plane $\text{PG}(d-1, N) = \text{PG}(N-1, N)$ is then self-referential: the field ($\mathbb{F}_N$) and the ambient dimension (d) share the same number. This self-referential structure exists only for $N = 3$.*

The decomposition $13 = 9 + 3 + 1 = N^2 + N + 1$ mirrors the gauge group $\text{U}(N) \times \text{SU}(2) \times \text{U}(1)$: $N^2 = \dim \text{U}(N)$, $N = \dim_{\text{fund}}$, $1 = \dim \text{U}(1)$.

5.2 Electroweak breaking as a geometric choice

In $\text{PG}(2, N)$, a *line* contains $N + 1 = 4$ points. Choosing a point P_0 (hypercharge) and a line L through it partitions the 13 points:

$$\underbrace{N^2}_{9 \text{ color}} + \underbrace{N}_{3 \text{ weak}} + \underbrace{1}_{1 \text{ hyp}} = N^2 + N + 1. \quad (5)$$

The $N = 3$ points on $L \setminus \{P_0\}$ are the weak directions; the $N^2 = 9$ points in $\text{PG} \setminus L$ are the color sector. Before symmetry breaking, all 13 directions are equivalent (GUT); the choice of (P_0, L) selects the Standard Model subgroup.

5.3 Gauge couplings

Gauge couplings from the invariant measure. The automorphism group $\text{PGL}(3, \mathbb{F}_3)$ acts *transitively* on $\text{PG}(2, 3)$: any point can be mapped to any other. The unique probability measure invariant under a transitive group action is uniform (Haar measure). Before symmetry breaking, the gauge-invariant weight of each direction is therefore $1/|\text{PG}| = 1/13$ —not a postulate but a theorem. The Weinberg angle [17] is the fraction of weak directions:

$$\sin^2 \theta_W = \frac{|\text{weak}|}{|\text{PG}|} = \frac{N}{N^2 + N + 1} = \frac{3}{13} = 0.2308. \quad (6)$$

W/Z mass ratio. The Weinberg angle determines the W/Z mass ratio at tree level:

$$\frac{m_W}{m_Z} = \cos \theta_W = \sqrt{1 - \frac{3}{13}} = \sqrt{\frac{10}{13}} = 0.8771 \quad (\text{experiment: } 0.8815, 0.5\%). \quad (7)$$

The 0.5% deviation is consistent with radiative corrections ($\Delta \rho \sim \alpha/\pi$).

Strong coupling from the torus. The automorphism group $\text{PGL}(d, \mathbb{F}_N)$ has maximal torus of dimension $\dim T = d - 1 = N - 1 = 2$. The tree-level strong coupling per torus direction is $\alpha_s^{(0)} = \sin^2 \theta_W / \dim T = 3/26 = 0.1154$. The $|\text{PG}| = N^2 + N + 1 = 13$ gauge directions decompose as $N = 3$ weak ($\text{SU}(2)$) and $N^2 + 1 = 10$ non-weak. The marginality correction acts through the $k = N^2 + 1 = 10$ non-weak PG directions:

$$\alpha_s = \frac{3}{26} \left(1 + \frac{\varepsilon}{N^2 + 1} \right) = 0.1179 \quad (\text{experiment: } 0.1179, 0.01\%). \quad (8)$$

Fine-structure constant from linear response. In 4D, $1/\alpha$ is determined by vacuum polarization—a closed fermion loop with two propagators ($\text{Tr}[G^2]$). In 0D there is no spacetime for pair creation; the gauge coupling is instead the *linear susceptibility* of the internal space to gauge field fluctuations:

$$\chi = - \left. \frac{\partial^2 F}{\partial A^2} \right|_{A=0} = \text{Tr} \left[\frac{1}{K_{\text{int}}} \right], \quad (9)$$

where $K_{\text{int}} = K_{\text{gen}} \otimes 1 + 1 \otimes \Delta_{\text{flag}}$ is the total internal kinetic operator (generation binding energies $|E_n|$ plus flag Laplacian λ_k). This is a theorem of linear response theory: the susceptibility is the resolvent trace ($s = 1$), not $\text{Tr}[1/K^2]$ ($s = 2$, which would give $1/\alpha \approx 9258$). The resolvent gives $Z = 134.78$ (the matter vacuum polarization Π_{matter}).

$\lambda_1 = 4 - \sqrt{3}$ is the bare photon inverse propagator D_{bare}^{-1} on the flag manifold (spectral gap = minimum gauge transition energy).

Derivation of the Dyson equation. The physical coupling $1/\alpha$ is the inverse dressed propagator, satisfying $D_{\text{dressed}}^{-1} = D_{\text{bare}}^{-1} + \Pi_{\text{matter}} = \lambda_1 + Z$. In 0D, the exact effective action $\Gamma = -H(\sigma(\phi))$ is known in closed form, and the Ward identity takes the form $\Gamma^{(3)} = \partial\Gamma^{(2)}/\partial\phi$, relating the vertex correction to the self-energy derivative. Since no loop integrals exist in 0D, there is no anomaly and Noether's theorem holds exactly. The vertex and self-energy corrections are therefore equal (the 0D specialization of $Z_1 = Z_2$), giving $\Pi = \alpha$. Explicitly: $\Gamma^{(2)}(\phi) = -V''(\phi) = H''(\sigma(\phi))$ and $\Gamma^{(3)}(\phi) = -V'''(\phi)$, so $\Gamma^{(3)} = \partial\Gamma^{(2)}/\partial\phi$ is an algebraic identity (not a dynamical Ward identity requiring regularization). The full derivation is given in Paper II [21]. Self-consistency $1/\alpha = (\lambda_1 + Z) - \alpha$ yields:

$$\frac{1}{\alpha} + \alpha = Z + \lambda_1. \quad (10)$$

Solution: $1/\alpha = [(Z + \lambda_1) + \sqrt{(Z + \lambda_1)^2 - 4}]/2 = 137.0367$ (experiment: 137.036, 0.0006%).

The resolvent trace decomposes as $Z = \zeta_V(1) + 12 S_1 + 12 S_2 + 27 S_3$, where $S_i = \sum_n (E_n + \lambda_i)^{-1}$ and the multiplicity $27 = N^N$ (Theorem 5). The N^N sector contributes $9.9 \approx N^2$, providing a spectral origin for the QCD vacuum-polarization correction. Only bound states enter; continuum states ($E \rightarrow 0^+$) are excluded because only physical generations contribute to vacuum polarization.

Experimental values: $\sin^2\theta_W = 0.2309$ at $\mu = v$ (0.06%), $\alpha_s(M_Z) = 0.1179$ (0.01%), $1/\alpha_{\text{em}} = 137.0367$ (0.0006%). All three gauge couplings are derived from $V = -H$ and $\text{PG}(2, 3)$.

5.4 Flag Laplacian spectrum

A flag in $\text{PG}(2, N)$ is a point-on-line incidence pair; flags are the natural basis for the gauge Laplacian because gauge fields propagate along incidence relations, not just points.

The flag variety of $\text{PG}(2, N)$ has $(N^2 + N + 1)(N + 1) = 52$ elements (point-line incidence pairs). The flag graph (adjacency when two flags share a point or line) has degree $2N = 6$ and Laplacian spectrum:

Eigenvalue	Multiplicity	
0	1	trivial
$(N+1) - \sqrt{N}$	$N^2 + N$	= 12
$(N+1) + \sqrt{N}$	$N^2 + N$	= 12
$2(N+1)$	N^N	= 27

The lowest non-trivial eigenvalue $\lambda_1 = (N+1) - \sqrt{N} = 4 - \sqrt{3} = 2.268$ matches $R_{\text{down}} = 2.272$ to 0.17%. The highest eigenvalue has multiplicity N^N , which equals $N^3 = 27$ only for $N = 3$ (Theorem 5). Whether the near-equality $\lambda_1 \approx R_{\text{down}}$ reflects a duality between the continuous spectrum of $V = -H$ and the discrete spectrum of the flag Laplacian is an open question.

5.5 Spectral participation ratio

The bound-state density $\rho(\phi) = \sum_{n=0}^{N-1} |\psi_n(\phi)|^2$ has participation ratio

$$D_{\text{PR}} = \frac{(\int \rho d\phi)^2}{\int \rho^2 d\phi} = \frac{N^2}{\sum_{nm} C_{nm}}, \quad C_{nm} = \int |\psi_n|^2 |\psi_m|^2 d\phi. \quad (11)$$

For $V = -H$ ($c = 1$): $D_{\text{PR}} = 13.06 = |\text{PG}(2, 3)|$ to 0.4%. Among the family $V = -cH$, $D_{\text{PR}}(c)$ is monotonically decreasing and passes through 13 near $c = 1$: $D_{\text{PR}}(0.95) = 13.8$, $D_{\text{PR}}(1.00) = 13.06$, $D_{\text{PR}}(1.05) = 12.4$. The derivation chain thus has two independent paths from $V = -H$ to $\text{PG}(2, 3)$: the eigenvalues determine $N = 3$ (Theorem 1); the eigenfunctions determine $D_{\text{PR}} \approx |\text{PG}| = 13$.

A look-elsewhere analysis over 2226 algebraic expressions from $\{N, N^2+N+1, N^2, N\pm 1, \pi, \pi^2\}$ gives joint probability $\lesssim 10^{-7}$ for the three gauge couplings matching simultaneously. Among integers $N = 2, \dots, 10$, only $N = 3$ reproduces all three (combined error 2.2%; next best $N = 4$ at 80%).

6 Generation as Quantized Occupation

Spin is the quantization of angular momentum: a continuous degree of freedom (rotation angle) yields discrete states ($\uparrow, \downarrow$) from $\text{SO}(3)$ topology.

Generation, in this framework, is the quantization of occupation probability: a continuous degree of freedom ($P \in (0, 1)$, or $\phi \in \mathbb{R}$) yields discrete states (τ, μ, e) from $V = -H$.

	Spin	Generation
Continuous variable	rotation angle θ	occupation $\phi = \text{logit}(P)$
Constraint	$\text{SO}(3)$ topology	$V = -H(\sigma(\phi))$
Discrete states	2 (for spin- $\frac{1}{2}$)	3 (Theorem 1)
Position-independent	✓	✓
Metric origin	round metric on S^2	Fisher metric on $(0, 1)$

Important difference. Spin quantization is topological ($\text{SO}(3)$: only integers and half-integers). Generation “quantization” is spectral (depends on the shape of $V = -H$). The number 3 is not topologically protected; it follows from the specific width-to-depth ratio of binary entropy.

Three dynamical regimes. The binding fraction $x_n = |E_n|/\langle H \rangle_n$ partitions the three modes into qualitatively distinct regimes: τ ($x = 0.85$, potential-dominated), μ ($x = 0.49$, equipartitioned), and e ($x = 0.12$, kinetic-dominated). The μ mode satisfies $\text{KL}(x_\mu \| \frac{1}{2}) = 0.0003 \approx 0$: its energy is partitioned at maximum entropy. $N = 3$ is not merely the count of half-wavelengths but the number of dynamical regimes that binary entropy supports. The electron’s stability follows from its delocalization: the shallowest mode has the smallest mass and no lighter charged lepton to decay into.

Koide relation [10] from equipartition. The Koide parameter $Q = (\sum \sqrt{m_n})^2 / \sum m_n$ equals $3 \cos^2 \alpha$, where α is the angle between the mass vector $v = (\sqrt{m_0}, \sqrt{m_1}, \sqrt{m_2})$ and the democratic direction $(1, 1, 1)$. $Q = 3/2$ iff $\cos^2 \alpha = 1/2$ iff $|v_\parallel|^2 = |v_\perp|^2$: the democratic component (equal masses) and the hierarchical component (mass differences) carry equal weight. This is the maximum-entropy condition for characterizing a mass spectrum as “democratic vs. hierarchical”—a binary question on a spectrum, directly reflecting A2.

The total fermion field formally decomposes as $\psi(x, \phi) = \sum_n \Psi_n(\phi) \cdot \chi_n(x)$, where Ψ_n are internal eigenfunctions and $\chi_n(x)$ are 4D Dirac fields—one per generation. This resembles Kaluza–Klein decomposition, but ϕ is not an extra spatial dimension: it is the Fisher-flat coordinate on the Bernoulli manifold, an information-geometric internal space. The “size” of this space is not a length but the range of occupation probability $P \in (0, 1)$. Whether this decomposition can be made dynamically consistent (internal and external

sectors decoupling) is addressed by the tensor product structure $\mathcal{H}_{\text{internal}}(\phi) \otimes \mathcal{H}_{\text{spacetime}}(x)$: the Schmidt decomposition is always possible and requires no dynamical justification (see Section 8.2).

7 Physical Correspondences

7.1 CKM structure [14]

Theorem 6 ($V_{ub} \approx 0$). *Parity: ψ_0 even, ψ_1 odd, ψ_2 even. $\int \psi_0 F \psi_2 d\phi = 0$. Numerically: 1.9×10^{-13} . ✓*

Screening identity: $E_2/\text{IPR}_2 = 0.209 \approx \varepsilon = 3 - n_{\text{WKB}} = 0.219$ (5%).

Cabibbo angle [13] from the WKB deficit. The WKB deficit $\varepsilon = N - n_{\text{WKB}}$ measures how marginal the third bound state is: $\varepsilon \rightarrow 0$ eliminates mode 2, reducing to two generations and zero CKM mixing. To second order:

$$\sin \theta_C = \varepsilon \left(1 + \frac{\varepsilon}{N^2} \right) = 0.2245 \quad (\text{experiment: } 0.2244, 0.05\%). \quad (12)$$

Both $N = 3$ and $n_{\text{WKB}} = 2.781$ are determined by $V = -H$; $N^2 = 9 = \dim \text{adj SU}(N)$. No free parameters enter. The Wolfenstein [16] hierarchy follows: $|V_{us}| \sim \varepsilon = 0.22$, $|V_{cb}| \sim \varepsilon^2 = 0.048$, (experiment: 0.225, 0.041, 0.004; correct hierarchy, each suppressed by $\sim \varepsilon$).

The NLO correction has k determined by the filtration structure of $\text{PGL}(3, \mathbb{F}_3)$: ascending corrections $(1 + \varepsilon/k)$ with $k = N^2 = 9$ for generation mixing ($\sin \theta_C$), $k = N^2 + 1 = 10$ for the non-weak sector (α_s); descending corrections $(1 - 2\varepsilon/k)$ with $k = N = 3$ for quantities suppressed at the center of the nilpotent radical ($\sin^2 \theta_{13}$). The decomposition $|\text{PG}| = N + (N^2 + 1) = 3 + 10 = 13$ separates pure-weak from non-weak directions.

Strong CP solution. The potential $V(\phi) = V(-\phi)$ is exactly CP-symmetric: $\phi \rightarrow -\phi$ corresponds to $P \rightarrow 1 - P$ (particle–antiparticle interchange in occupation space). Since the fundamental Lagrangian inherits this symmetry, $\theta_{\text{QCD}} = 0$ identically—the strong CP problem does not arise. No axion is needed. Experiment: $|\theta_{\text{QCD}}| < 10^{-10}$. Prediction: $\theta_{\text{QCD}} = 0$ (exact). ✓

CP violation in the CKM sector. $V(\phi) = V(-\phi)$ implies exact CP at leading order ($V_{ub} = 0$, Jarlskog $J = 0$). The observed CKM CP violation ($J = 3.18 \times 10^{-5}$) arises at higher order: the Wolfenstein structure $J \propto \varepsilon^6$ gives $J \sim 10^{-4}$, the correct order of magnitude. Strong CP (θ_{QCD}) is protected by the *exact* $\phi \leftrightarrow -\phi$ symmetry of $V = -H$.

7.2 Mass formula

The mass formula is derived from the self-energy structure of the 0D theory. The physical mass is $m_n = |E_n| \cdot e^{\Sigma(n)}$, where $\Sigma(n)$ is the self-energy. Two vertices contribute: the gauge vertex ($\bar{\psi}\psi$ bilinear) and the potential vertex ($V = -H$). Their contributions are additive: $\Sigma(n) = \Sigma_{\text{gauge}}(n) + \Sigma_{\text{pot}}(n)$. Exponentiation gives a product:

$$m_n = |E_n| \times e^{\Sigma_{\text{gauge}}} \times e^{\Sigma_{\text{pot}}}. \quad (13)$$

The product structure is not an ansatz but the exponential of an additive self-energy.

Gauge vertex: Callan–Symanzik equation. In 0D, the inverse participation ratio $\text{IPR}_n = \int |\psi_n|^4 d\phi$ is the only available scale (localized states couple more strongly

to gauge fields). In 0D there is no momentum to “run,” but the RG operates across eigenstates: different modes have different IPR (localization), and the mass–IPR scaling is the content of the renormalization group. The 0D Callan–Symanzik equation $[\text{IPR} \partial/\partial \text{IPR} + \gamma_m] m = 0$ with $\gamma_m = 2C_F = (N^2 - 1)/N = 8/3$ (two fermion legs, each carrying C_F ; no loop integral in $d = 0$, so the exponent is pure group theory) has solution $\Sigma_{\text{gauge}}(n) = 2C_F \ln \text{IPR}_n$.

Potential vertex: eigenstate-dependent coupling. The potential $V = -H$ generates a second self-energy insertion proportional to the eigenstate’s entropy overlap. The relevant eigenstate-dependent quantity is $\langle H \rangle_n / T_n$, the ratio of potential to kinetic expectation values—the virial ratio, which measures how each mode samples the anharmonicity of $V = -H$. For a harmonic potential (virial exact), $\langle H \rangle / T = 1$ and the correction vanishes. Hence $\Sigma_{\text{pot}}(n) = \delta \ln(\langle H \rangle_n / T_n)$.

Spectral ζ -function and the exponent δ . The spectral ζ -function $\zeta_V(s) = \sum_n |E_n|^{-s}$ has WKB residue $\text{Res}_{s=1} \zeta_V = n_{\text{WKB}} = 2.781$. Each of the $N = 3$ exact modes “should” contribute 1 to this residue; the deficit per mode is $\delta = (N - n_{\text{WKB}})/N = \varepsilon/N = 0.073$. This is the anomalous dimension of the potential vertex: each mode undercontributes to the spectral density by a fraction ε/N , and this deficit appears as the exponent of the virial correction.

Combining:

$$m_n \propto |E_n| \times \text{IPR}_n^{2C_F} \times \left(\frac{\langle H \rangle_n}{T_n} \right)^{\varepsilon/N}, \quad (14)$$

where $T_n = \frac{1}{2} \int \psi_n'^2 d\phi$ (gradient energy—a wavefunction property, not the BenDaniel–Duke [15] kinetic energy), $\langle H \rangle_n = \int \psi_n^2 H d\phi$ (entropy expectation). Every element is derived: the product structure from self-energy additivity, $2C_F$ from the Callan–Symanzik equation, $\langle H \rangle / T$ from the potential vertex, and ε/N from the spectral ζ -function. No free parameters remain.

Verification: Koide $Q = 3/2$. The Koide parameter $Q = 3/2$ (Section 6) is not an input but a consequence. Given $b = 2C_F$ and $c = \varepsilon/N$, the formula yields $Q = 1.500$ exactly. This is independently verified: no other pair (b, c) simultaneously satisfies $Q = 3/2$ and $c = \varepsilon/N$:

b	$c(Q=3/2)$	$ c - \varepsilon/N /(\varepsilon/N)$
2.0	0.437	500%
2.5	0.163	120%
8/3 = 2C_F	0.0730	0.06%
2.78	0.011	85%
3.0	−0.107	246%

The exponents are not fitted to mass data; $R = 1.5293$ is a *prediction*.

	Eq. (14)	Experiment	Accuracy
m_τ/m_e	3480	3477	0.1%
m_μ/m_e	207	207	0.0%
R	1.5293	1.5294	0.006%
Q (Koide)	1.500	1.500	< 0.01%

$R \in [1.44, 1.57]$ for all power-law forms: a parameter-free bound.

Convergence. All spectral quantities—eigenvalues, IPR, virial ratios, and the resolvent trace (10)—are computed with $|\phi|_{\text{max}} = 60$, $\Delta\phi = 0.0002$ ($N_{\text{grid}} = 512001$), and

verified stable under doubling of grid resolution ($|\psi_{\text{edge}}| < 10^{-9}$). At $|\phi|_{\text{max}} = 25$: E_2 is 0.8% too low, R degrades to 1.536 (0.4%), and $1/\alpha_{\text{em}}$ shifts to 137.78 (0.5%). At $|\phi|_{\text{max}} = 100$: $R = 1.529$ and $1/\alpha_{\text{em}} = 137.035$, confirming convergence at $|\phi|_{\text{max}} = 60$.

7.3 Up-type quarks: $c = N_c = 3$

The color number $N_c = 3$ is derived, not assumed: $d = 3$ (Theorem 4) $\rightarrow \nabla\phi \in \mathbb{C}^3 \rightarrow \text{CP}^2 \rightarrow \text{SU}(3)_{\text{color}} \rightarrow N_c = 3$. Color confinement forces all N_c color modes to share a single source field ϕ (because confined quarks cannot be isolated; their occupation parameters are locked together). In the 0D QFT, N_c identical modes give

$$Z_{N_c}(\phi) = (1 + e^\phi)^{N_c}, \quad W_{N_c} = N_c \ln(1 + e^\phi), \quad \Gamma_{N_c} = N_c \Gamma_1 = -N_c H(\sigma). \quad (15)$$

Hence $V = -N_c H = -cH$ with $c = N_c$: the potential depth is determined by the color number, not fitted. At $c = 3$: $N = 5$ bound states, $n_{\text{WKB}} = 4.82$.

With the same exponents $b = 2C_F$, $c = \varepsilon/N$ as the lepton sector (§6.2) and modes $(1, 2, 3)$: $R_{\text{up}} = 1.777$ (experiment: 1.772, deviation 0.3%). Both $c = 3$ and $b = 2C_F$ are derived quantities. This is a zero-parameter prediction.

7.4 Down-type quarks: effective mass from weak isospin

At $c = 3$ with constant mass, modes $(0, 1, 2)$ give $R_{\text{down}} = 2.096$ (experiment: 2.269, 7.6% off). The discrepancy is resolved by a variable effective mass.

In the symmetric sector $\phi_r = \phi_g = \phi_b = \phi$, fluctuations off the diagonal induce an effective mass correction proportional to the Fisher metric $g_F = \sigma(1 - \sigma)$, which is maximal at $P = 1/2$ (maximum uncertainty) and vanishes at $P = 0, 1$:

$$m_{\text{eff}}(\phi) = 1 + g \sigma(1 - \sigma). \quad (16)$$

The coupling g depends on weak isospin I_3 :

$$g = 0 \quad (I_3 = +\frac{1}{2}), \quad g = -g_0 \quad (I_3 = -\frac{1}{2}), \quad (17)$$

with

$$g_0 = 2C_F + \frac{1}{N_c} = \frac{N_c^2 - 1}{N_c} + \frac{1}{N_c} = \frac{N_c^2}{N_c} = N_c = 3. \quad (18)$$

The vanishing of g for $I_3 = +1/2$ ensures that the up-quark R -value remains parameter-free. The identity $2C_F + 1/N_c = N_c$ is exact: $2C_F = (N_c^2 - 1)/N_c$ is the SU(2) doublet's total color Casimir, and $1/N_c$ is the baryon U(1) charge. The $I_3 = -1/2$ assignment follows from three steps: (1) the algebraic identity $2C_F + 1/N_c = N_c$ holds for all N_c ; (2) A2 (Contrast) requires the doublet's full charge to reside in the contrast component ($I_3 = -1/2$), since $I_3 = +1/2$ is the reference; (3) the alternative assignment $g = -N_c/2$ degrades R_{down} from 0.2% to 7.6%, quantitatively excluding it. Numerical verification: $R_{\text{down}} = 2.273$ (experiment: 2.269, 0.2%).

With $g = -N_c = -3$ and modes $(0, 1, 2)$: $R_{\text{down}} = 2.273$ (experiment: 2.269, deviation 0.2%).

Sector	c	g	R (predicted)	R (experiment)
Lepton	1	0	1.529	1.529 (0.006%)
Up	$N_c = 3$	0	1.777	1.772 (0.3%)
Down	$N_c = 3$	$-N_c$	2.273	2.269 (0.2%)

All three R -values are reproduced within $\sim 1\%$. All ingredients are derived: $c = N_c = d = 3$ (Theorem 4 $\rightarrow$ SU(3) $\rightarrow N_c = 3$; confinement $\rightarrow c = N_c$); $g_0 = 2C_F + 1/N_c = N_c$ (doublet Casimir + baryon U(1)); mode assignments up = (1, 2, 3), down = (0, 1, 2) (at $c = N_c$, mode 0 is deeply confined below the generation window; the contrast component with $g = -N_c$ has $N = 4$ states, and (0, 1, 2) are the three lightest).

Derivation of g_0 . The SU(2) doublet (u, d) carries total color Casimir $2C_F = (N_c^2 - 1)/N_c = 8/3$, plus baryon U(1) charge $1/N_c = 1/3$. The A2 contrast structure motivates assigning the *entire* doublet charge to the $I_3 = -1/2$ component:

$$g_0 = 2C_F + \frac{1}{N_c} = \frac{N_c^2 - 1}{N_c} + \frac{1}{N_c} = \frac{N_c^2}{N_c} = N_c = 3. \quad (19)$$

This identity is exact for all N_c . The result $g_0 = c = N_c$ means a single quantity—the number of confined color modes—determines both the potential depth ($V = -N_c H$) and the mass correction ($m_{\text{eff}} = 1 - N_c \sigma(1 - \sigma)$).

7.5 Neutrino mass prediction

The ratio R is universal for all fermions coupled to $V = -H$ with $c = 1$: it depends only on the eigenstate properties of the potential, not on the absolute mass scale. Neutrinos are color-singlets and therefore couple to $V = -H$ with $c = 1$ (the same potential as charged leptons); the mass ratio $R_\nu = R_{\text{lepton}} = 1.529$ follows. The absolute neutrino mass scale requires spacetime input (Yukawa couplings), but the ratio is determined internally. Combined with oscillation data ($\Delta m_{21}^2 = 7.53 \times 10^{-5} \text{ eV}^2$, $\Delta m_{31}^2 = 2.45 \times 10^{-3} \text{ eV}^2$, normal ordering), $R_\nu = 1.529$ determines m_1 uniquely:

	Prediction	Status
m_1	$0.32 \pm 0.03 \text{ meV}$	unmeasured
m_2	8.68 meV	consistent
m_3	49.5 meV	consistent
$\sum m_i$	$58.55 \pm 0.3 \text{ meV}$	below Planck (< 120)
Ordering	normal	JUNO (2026–27)

The sum $\sum m_i = 58.55 \text{ meV}$ is dominated by $m_3 \approx \sqrt{\Delta m_{31}^2} = 49.5 \text{ meV}$; the theory's contribution is $m_1 \approx 0.3 \text{ meV}$, placing $\sum m_i$ near its minimum allowed value. The Koide parameter $Q_\nu = 1.90 \neq 3/2$: neutrinos do not satisfy the equipartition condition, consistent with the theory's prediction that $Q = N/2$ holds only for charged leptons (all $N = 3$ modes used).

Falsifiability. (i) Inverted ordering $\rightarrow$ theory falsified. (ii) $\sum m_i > 70 \text{ meV}$ (requiring $R > 1.8$) $\rightarrow$ tension. (iii) CMB-S4 sensitivity ($\sigma \sim 15 \text{ meV}$) can test $\sum m_i = 58.55 \text{ meV}$ directly.

7.6 Higgs mass ratio

The potential $V = -H$ has curvature at the center: $V''(0) = g_F(0) = \sigma(0)(1 - \sigma(0)) = 1/4$, the Fisher metric at maximum uncertainty ($P = 1/2$). In QFT, $m^2 = V''(\phi_{\text{min}})$ gives the mass of small oscillations:

$$\left. \frac{m_H}{v} \right|_{\text{LO}} = \sqrt{V''(0)} = \frac{1}{2}. \quad (20)$$

The NLO correction from the PG(2,3) gauge structure:

$$\frac{m_H}{v} = \frac{1}{2} \sqrt{1 + \frac{2\varepsilon}{|\text{PG}|}} = \frac{1}{2} \sqrt{1 + \frac{2 \times 0.219}{13}} = 0.5084 \quad (\text{experiment: } 0.5087, 0.07\%). \quad (21)$$

This corresponds to $\lambda = (m_H/v)^2/2 = 0.1292$ (experiment: 0.1294, 0.11%).

7.7 Neutrino mixing angles

CKM mixing is perturbative ($\sin \theta_C \approx \varepsilon \ll 1$): the internal coordinate ϕ controls mixing. PMNS mixing is non-perturbative ($\sin^2 \theta_{12} \approx 1/3$): the gauge geometry controls mixing.

A line in PG(2,3) contains $N+1 = 4$ points. The solar mixing angle measures the fraction of PG covered by one complete electroweak line ($\text{SU}(2) \times \text{U}(1)$, including hypercharge):

$$\sin^2 \theta_{12} = \frac{N+1}{|\text{PG}|} = \frac{4}{13} = 0.3077 \quad (\text{experiment: } 0.307, 0.2\%). \quad (22)$$

Compare $\sin^2 \theta_W = N/|\text{PG}| = 3/13$ (pure SU(2), excluding hypercharge); the difference $1/|\text{PG}|$ is the $\text{U}(1)_Y$ contribution.

The atmospheric mixing is maximal (μ - τ symmetry) plus a correction at the PG scale:

$$\sin^2 \theta_{23} = \frac{1}{2} + \frac{1}{2|\text{PG}|} = \frac{7}{13} = 0.538 \quad (\text{LO; experiment: } 0.546, 1.4\%). \quad (23)$$

The NLO Borel correction with $k = |\text{PG}| = 13$:

$$\sin^2 \theta_{23} = \frac{7}{13} \left(1 + \frac{\varepsilon}{13}\right) = 0.5475 \quad (\text{experiment: } 0.546, 0.3\%). \quad (24)$$

The correction $1/(2|\text{PG}|)$ is the minimum μ - τ symmetry breaking from the gauge geometry; the sign ($> 1/2$) follows from τ being heavier (more localized, larger PG overlap). Consistency check: $\sin^2 \theta_{23} = \sin^2 \theta_W + \sin^2 \theta_{12}$ ($7/13 = 3/13 + 4/13$): atmospheric mixing exhausts both the pure-weak (N points) and the full electroweak-line ($N+1$ points) fractions of PG(2,3). The reactor angle is suppressed by the full PG: $\sin^2 \theta_{13} = 1/(N \cdot |\text{PG}|) = 1/39 = 0.026$ at leading order. The NLO Borel correction at the center $Z(U)$ ($k = N = 3$) gives $\sin^2 \theta_{13} = (1/39)(1 - 2\varepsilon/3) = 0.0219$ (experiment: 0.022, 0.5%). NLO corrections to all three PMNS angles and the CKM CP phase are derived in Paper II [21].

Status of PMNS. CKM involves the internal coordinate ϕ ($\varepsilon \ll 1$, perturbative); PMNS involves the gauge geometry ($1/N \sim 1$, non-perturbative). $\theta_{12} = 4/13$ is a PG line fraction (Tier B); $\theta_{23} = 7/13$ follows from μ - τ breaking at the PG scale (Tier B).

7.8 Neutrinos are Dirac particles

Two independent arguments establish that neutrinos are Dirac, not Majorana:

(1) Grassmann antisymmetry. Fermion creation operators satisfy $c^\dagger c^\dagger = 0$ (Pauli exclusion). A Majorana mass term $\nu^T C \nu$ requires two identical fermion fields, violating this antisymmetry in the single-mode ($n \in \{0, 1\}$) framework.

(2) V-parity. $V(\phi) = V(-\phi)$: the potential is symmetric under $\phi \rightarrow -\phi$ ($P \rightarrow 1-P$). This Z_2 symmetry forbids all lepton-number-violating operators, including Majorana mass terms.

Prediction: neutrinoless double beta decay is absent ($0\nu\beta\beta = 0$), testable by nEXO (~ 2030).

8 Connection to Four-Dimensional Physics

8.1 Two geometries from $H(P)$

Internal geometry. $H(P) \rightarrow V = -H$ (canonical free energy); $\phi = \text{logit}(P)$ (natural parameter) $\rightarrow$ Schrödinger equation.

Spacetime (Einstein). The action $S[\phi, g_{\mu\nu}]$ depends on the metric $g_{\mu\nu}$. Background independence requires $g_{\mu\nu}$ to be dynamical: fixing $g_{\mu\nu} = \eta_{\mu\nu}$ would be a structural choice not determined by the theory. Lovelock’s theorem [8] then guarantees that Einstein’s equation $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$ is the *unique* gravitational dynamics in $d+1 = 4$ dimensions (in $d+1 \geq 6$, Gauss–Bonnet terms appear; the derivation of $d = 3$ (Theorem 4) eliminates them). Independently, Jacobson [7]: $\delta Q = T dS$ on causal horizons $\rightarrow$ Einstein’s equation. If $S = H(P)$ fundamentally, both routes—action-based (Lovelock) and thermodynamic (Jacobson)—yield Einstein gravity from binary entropy.

8.2 Internal–external separation: tensor product structure

The decomposition $\Psi(x, \phi) = \sum_n \psi_n(\phi) \chi_n(x)$ is not a dynamical mechanism but a consequence of the tensor product structure $\mathcal{H}_{\text{total}} = \mathcal{H}_{\text{internal}}(\phi) \otimes \mathcal{H}_{\text{spacetime}}(x)$. The information coordinate $\phi = \text{logit}(P)$ is not a spatial direction, just as spin is not a spatial direction. Three independent arguments support this:

1. *Generation universality.* All gauge couplings are generation-independent, implying zero ϕ – x coupling: $H_{\text{int}} = H_{\text{gauge}}(x) \otimes \mathbf{1}_\phi$.
2. *Dimensionless–dimensional separation.* The 0D theory computes dimensionless quantities (mass ratios, mixing angles, $\sin^2 \theta_W$). Dimensional quantities (absolute masses, Λ_{QCD}) require spacetime input. This is exactly what a tensor product predicts: the internal factor determines ratios, the spacetime factor supplies units.
3. *5D embedding fails.* Promoting ϕ to a spatial coordinate (domain-wall mechanism [11]) yields an effective potential $V_L = H^2 + \phi\sigma(1-\sigma) \neq \frac{1}{4} + V$, which does not reproduce the 0D spectrum. $N = 3$ is lost. This failure confirms that ϕ is categorically distinct from spatial coordinates.

The Schmidt decomposition $\Psi = \sum_n \psi_n(\phi) \cdot \chi_n(x)$ is always possible in a tensor product and requires no dynamical justification.

9 Results and Discussion

The derivation chain from A1–A3 to the Schrödinger equation involves four steps, each with independent justification: (i) $V = \Gamma = -H$ is the exact 0D effective potential (Legendre transform); (ii) $Z_2 = 1$ follows from the flat path-integral measure in ϕ (exponential-family structure, not a choice); (iii) the transfer matrix of the canonical action *is* the Schrödinger equation (standard lattice QFT); (iv) $d = 3$ is the unique dimension preserving $N = 3$ with spin-statistics and propagating gravity. The identification $\mathbb{F}_3$ (not $\mathbb{C}^3$) follows from non-degeneracy (Sturm–Liouville $\rightarrow$ Schur $\rightarrow$ permutations $\rightarrow N = 3$ prime $\rightarrow \mathbb{F}_3$ unique); $D_{\text{PR}} = 13.06 \approx |\text{PG}(2, 3)|$ provides independent evidence. Self-consistency: $N = 3$ describes energy *levels* of one mode ($Z = 1 + e^\phi$), not three independent fields.

The quark sector introduces one physical input (confinement: $\phi_r = \phi_g = \phi_b$) yielding $c = N_c = 3$ and $R_{\text{up}} = 1.777$ (0.3%). The A2 contrast structure motivates $g = 0$ ($I_3 = +1/2$) and $g = -N_c$ ($I_3 = -1/2$), giving $R_{\text{down}} = 2.273$ (0.2%). The mass formula

exponents are determined by two constraints: $2C_F$ (bilinear $\bar{\psi}\psi$) and ε/N (Koide $Q = N/2$); no other IPR exponent satisfies both. Internal–external separation $\Psi(x, \phi) = \sum_n \psi_n(\phi) \chi_n(x)$ follows from the tensor product $\mathcal{H}_{\text{internal}} \otimes \mathcal{H}_{\text{spacetime}}$ (Section 8.2): ϕ is an information coordinate, not a spatial coordinate.

The 0D theory computes dimensionless internal-structure parameters in three categories: (A) scale-independent quantities (mass ratios, mixing angles, Q , θ_{QCD}); (B) the IR coupling $1/\alpha = 137.036$ (resolvent at $q^2 = 0$, Thomson limit); (C) tree-level gauge ratios ($\sin^2\theta_W = 3/13$, α_s), where $3/13$ is a counting fraction on $\text{PG}(2, 3)$ with no intrinsic scale. At the electroweak VEV scale ($\mu = v = 246$ GeV), $\sin^2\theta_W(\mu=v) = 0.2309$, matching $3/13 = 0.2308$ to 0.06%. At M_Z : $\sin^2\theta_W = 0.2312$ (0.2%); at $q^2 = 0$: $\sin^2\theta_W = 0.2389$ (3.4%). The PG structure matches the SM at the electroweak scale. Spacetime enters only through RG running.

PFE should not be confused with Fisher-information approaches to physics [9] (for criticism see [20]). In Extreme Physical Information (EPI), physical laws are *postulated* to extremize Fisher information—an additional variational principle. In PFE, $V = -H$ is the Legendre transform of the partition function, an algebraic identity requiring no variational postulate. The Fisher metric enters only through the kinetic term (§2), where it is fixed by the exponential-family structure, not extremized.

PFE also differs fundamentally from numerological coincidences (Eddington’s $\alpha^{-1} = 136$; Wyler’s $\alpha^{-1} = (9/16\pi^3)(2\pi^5/15)^{1/4} = 137.036$; Lenz’s $m_p/m_e = 6\pi^5 = 1836.1$). These are isolated numerical matches with no derivational structure, no second prediction, and no falsifiability. PFE derives 20+ quantities *simultaneously* from one equation with zero parameters. The simultaneous fit is the decisive distinction: a single numerological match has probability $\sim 1/137$; twenty simultaneous matches at sub-percent accuracy have probability $< 10^{-40}$ under the null hypothesis of independent coincidences.

Tier A: Mathematical theorems (from $V = -H$ alone).

Quantity	Prediction	Experiment	Proof
N (generations)	3	3	Variational + Sturm
d (spatial dim.)	3	3	Centrifugal + fixed point
Q (Koide)	$3/2$	$3/2$	Equipartition (A2)
θ_{QCD}	0	$< 10^{-10}$	$V(\phi) = V(-\phi)$
$V_{ub} \approx 0$	10^{-13}	0.004	Parity selection
4th generation	forbidden	excluded	$E_3 > 0$

Tier B: QFT derivations ($V = -H + \text{PG}(2, 3) + \text{standard QFT}$).

Quantity	Prediction	Experiment	Accuracy
$1/\alpha_{\text{em}}$	137.0367	137.036	0.0006%
R_{lepton}	1.5293	1.5294	0.006%
α_s	0.1179	0.1179	0.01%
$\sin \theta_C$	0.2245	0.2244	0.05%
m_τ/m_e	3480	3477	0.1%
m_μ/m_e	207	207	0.0%
R_{up}	1.777	1.772	0.3%
R_{down}	2.273	2.269	0.2%
$\sin^2 \theta_W$	$3/13$	0.2309	0.06%
m_W/m_Z	$\sqrt{10/13}$	0.8815	0.5%
$\sin^2 \theta_{12}$	$4/13$	0.307	0.2%
$\sin^2 \theta_{23}$	0.5475	0.546	0.3%
$\sin^2 \theta_{13}$	0.0219	0.0220	0.5%
m_H/v	0.5084	0.5087	0.07%

Predictions (testable, unconfirmed).

Quantity	Prediction	Test
Mass ordering	Normal	JUNO (2026–27)
Σm_ν	58.55 meV	CMB-S4 (~ 2030)
m_1	0.32 meV	Cosmological constraints
$0\nu\beta\beta$	Strictly zero	nEXO (~ 2030)

All predictions follow from one potential ($V = -H$) and one geometric structure ($\text{PG}(2, 3)$). Of the four predictions, mass ordering, m_1 , and $0\nu\beta\beta$ are independent; Σm_ν follows from m_1 and oscillation data. Further NLO corrections are derived in Paper II [21].

10 Conclusion

This paper has shown that a single equation $V = -H(\sigma(\phi))$, derived from three axioms, determines 20 physical quantities of the Standard Model with zero free parameters. Every prediction is individually falsifiable; the nearest test is the determination of neutrino mass ordering by JUNO (2026–27).

A rigorous proof of why this equation—and no other—follows necessarily from Fermi statistics, the Legendre transform, and Čencov’s theorem is given in Paper II. The cosmological consequences of $V = -H$ (Planck mass, cosmological constant, inflation, dark matter, baryogenesis, gravitational waves) are developed in Paper III.

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